

## Atomics

- Allow well-defined race conditions
- Allow fine-grained synchronization between threads

### Read-modify-write (RMW) operations

An RMW operation on an atomic variable: Reads the old value of the variable. Writes a new value to the variable. The new value is either a completely new value or a function of the old value

All this is performed as one indivisible step - other threads cannot observe the effects of an RMW operation midway through.

### **std::atomic<T>:**

- Declares an atomic variable of type T
- Most common use cases: boolean, integral and pointer types

```
std::atomic<MyStruct *> p(&my_struct);
```

Operations on std::atomic<T>:(All operations are atomic)

*store(x)*: store value x of type T

*load()*: yields value of type T